Taller 1, puntos 11 y 12: Adaline sobre las particiones de banknote

```
clear; close all; clc
load banknote_splits.mat
```

Parametros

```
umbral = 0.5; salida = [0 1]; error_min = 1e-3; max_epocas = 500; semilla = 1;
alphas = [0.0001 0.001 0.005 0.01 0.05];
```

Entrenamiento con y sin escalar

Sin escalar, la curtosis llega a 17 y con alpha grande el error se dispara

```
R = table();
for escalar = [false true]
    for k = 1:4
        Xtr = S(k).Xtr; Xte = S(k).Xte; dtr = S(k).ytr; dte = S(k).yte;
        if escalar
            mn = min(Xtr); mx = max(Xtr);
            Xtr = (Xtr - mn) ./ (mx - mn);  Xte = (Xte - mn) ./ (mx - mn);
        end
        for a = alphas
            [w, b, ep, EC] = adaline(Xtr, dtr, a, error_min, max_epocas, semilla);
            acc_tr = 100*mean(predecir(Xtr, w, b, umbral, salida) == dtr);
            acc_te = 100*mean(predecir(Xte, w, b, umbral, salida) == dte);
            R = [R; table(escalar, k, {sprintf('%d-%d', 100*S(k).p,
100-100*S(k).p)}, a, ep, EC(end), acc_tr, acc_te, ...
                 'VariableNames',
{'Escalado','k','Particion','Alpha','Epocas','EC_final','Exact_train','Exact_test'})
];
        end
    end
end
R
```

```
R = 40×8 table
```

...

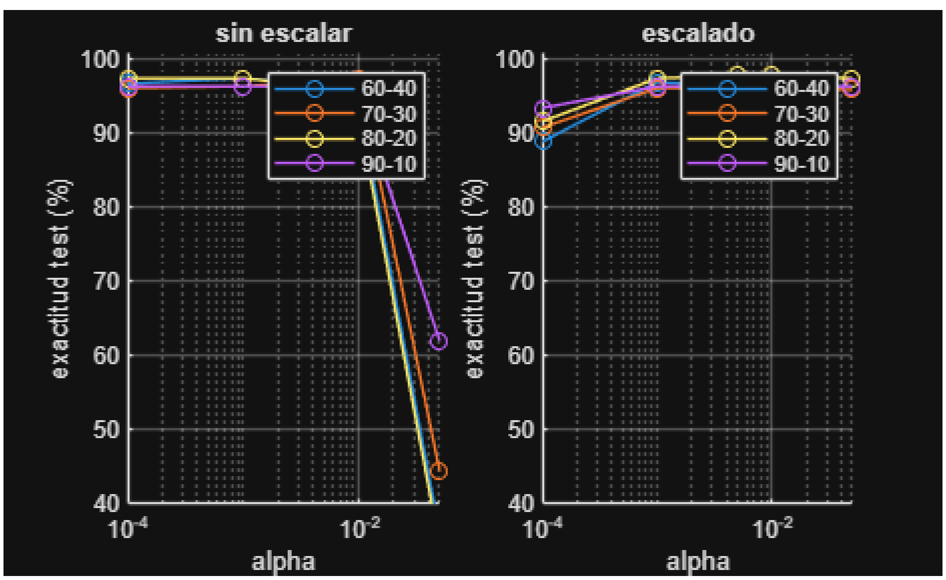
	Escalado	k	Particion	Alpha	Epocas	EC_final	Exact_train
1	false	1	'60-40'	1.0000e-04	225	13.1766	97.9344
2	false	1	'60-40'	1.0000e-03	41	13.9384	98.0559
3	false	1	'60-40'	0.0050	11	18.9252	97.5699
4	false	1	'60-40'	0.0100	6	39.0029	95.3827
5	false	1	'60-40'	0.0500	2	Inf	37.1810
6	false	2	'70-30'	1.0000e-04	199	15.2026	98.0208
7	false	2	'70-30'	1.0000e-03	36	16.0786	98.5417

	Escalado	k	Particion	Alpha	Epocas	EC_final	Exact_train
8	false	2	'70-30'	0.0050	10	21.9613	98.7500
9	false	2	'70-30'	0.0100	6	46.7720	98.3333
10	false	2	'70-30'	0.0500	2	Inf	44.4792
11	false	3	'80-20'	1.0000e-04	180	18.7541	97.9964
12	false	3	'80-20'	1.0000e-03	31	19.7118	97.5410
13	false	3	'80-20'	0.0050	8	26.4692	95.8106
14	false	3	'80-20'	0.0100	5	56.0447	93.7158

⋮

Grafica

```
figure
titulos = {'sin escalar', 'escalado'};
for e = 1:2
    subplot(1,2,e); hold on
    for k = 1:4
        m = R.Escalado == (e==2) & R.k == k;
        plot(R.Alpha(m), R.Exact_test(m), '-o', 'DisplayName',
R.Particion{find(m,1)});
    end
    set(gca, 'XScale', 'log'); grid on; ylim([40 101]); legend
    xlabel('alpha'); ylabel('exactitud test (%)'); title(titulos{e});
end
```



Mejor caso: curva de error y matriz de confusion

```
[~, i] = max(R.Exact_test);
```

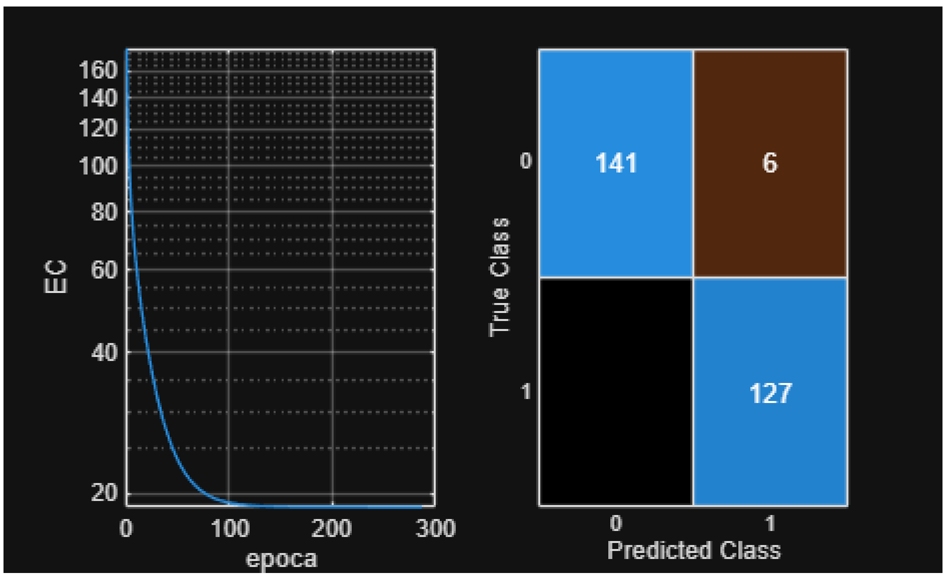
```
R(i, :)
```

ans = 1×8 table

...

	Escalado	k	Particion	Alpha	Epocas	EC_final	Exact_train
1	true	3	'80-20'	0.0050	287	18.7239	98.0874

```
k = R.k(i);
Xtr = S(k).Xtr; Xte = S(k).Xte;
if R.Escalado(i)
    mn = min(Xtr); mx = max(Xtr);
    Xtr = (Xtr - mn) ./ (mx - mn);  Xte = (Xte - mn) ./ (mx - mn);
end
[w, b, ~, EC] = adaline(Xtr, S(k).ytr, R.Alpha(i), error_min, max_epocas, semilla);
figure
subplot(1,2,1); semilogy(EC); grid on; xlabel('epoca'); ylabel('EC')
subplot(1,2,2); confusionchart(S(k).yte, predecir(Xte, w, b, umbral, salida))
```



Perceptron contra Adaline, particion 80-20 escalada

```
Xtr = S(3).Xtr; Xte = S(3).Xte;
mn = min(Xtr); mx = max(Xtr);
Xtr = (Xtr - mn) ./ (mx - mn);  Xte = (Xte - mn) ./ (mx - mn);
R2 = table();
for a = [0.001 0.01 0.05 0.1]
    [wp, bp, ep_p] = perceptron(Xtr, S(3).ytr, 3, a, 0.5, [0 1], 500, 1);
    [wa, ba, ep_a] = adaline(Xtr, S(3).ytr, a, error_min, max_epocas, 1);
    R2 = [R2; table(a, ep_p, 100*mean(predecir(Xte,wp,bp,0.5,[0 1]) == S(3).yte),
...
                    ep_a, 100*mean(predecir(Xte,wa,ba,0.5,[0 1]) == S(3).yte), ...
          'VariableNames',
{'Alpha','Epocas_perc','Exact_perc','Epocas_ada','Exact_ada'})];
end
```

```
R2
```

R2 = 4×5 table

	Alpha	Epocas_perc	Exact_perc	Epocas_ada	Exact_ada
1	1.0000e-03	500	98.9051	500	97.4453
2	0.0100	500	98.9051	154	97.8102
3	0.0500	500	98.9051	46	97.4453
4	0.1000	500	97.4453	26	97.4453